OpenReview.net(/)

← Go to **NeurIPS 2026 Conference** homepage (/group?id=NeurIPS.cc/2026/Conference)

Bayesian Optimization on Function Spaces via Sparse RKHS Manifolds

(/pdf?id=hBh55r5Asy)

Davide Sartor (/profile?id=~Davide_Sartor1), Meghan E. Huber (/profile?id=~Meghan_E._Huber1), Donghyun Kim (/profile?id=~Donghyun_Kim11), Nathan Wycoff (/profile?id=~Nathan_Wycoff2)

Published: 24 Sept 2026, Last Modified: 24 Sept 2026 NeurIPS 2026 poster
Conference, Senior Area Chairs, Area Chairs, Reviewers, Authors
Revisions (/revisions?id=hBh55r5Asy) BibTeX
CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/)

TL;DR: A new Functional BO method searching sparse RKHS sub-manifolds, jointly optimizing coefficients and support locations. Ships the first open-source FBO benchmark suite (70+ functional problems) and beats existing baselines on synthetic and real tasks.

Abstract:
Bayesian Optimization has become an established methodology for minimizing black-box functions of a vector input. Often, however, this parameter vector arises from the discretization of an inherently functional relationship. Several recent articles have considered the Functional Bayesian Optimization (FBO) setting, in which the variable to be optimized is not a member of a finite dimensional vector space, but rather an infinite dimensional function space. In this work, we propose L^0 Manifold Optimization (L0MO), a simple approach to FBO which searches a sub-manifold of a Reproducing Kernel Hilbert Space consisting of functions with a sparse representation in the kernel functions. This results in simpler, more effective functional solutions than existing methods. We discuss in detail the relationship between our method and existing ones, providing a unifying lens through which to view prior works. To assess our method against the state of the art, we conduct an extensive computational study, and along the way develop a novel set of benchmark test functions which port standard finite-dimensional ones to the infinite dimensional domain. Our experiments demonstrate that the proposed method achieves superior performance across a wide range of test benchmarks.

Checklist Confirmation: I confirm that I have included a paper checklist in the paper PDF.
Supplementary Material: zip (/attachment?id=hBh55r5Asy&name=supplementary_material)
Responsible Reviewing: We acknowledge the responsible reviewing obligations as authors.
Primary Area: Probabilistic methods (e.g., variational inference, causal inference, Gaussian processes, sampling)
Secondary Area: Optimization (e.g., convex and non-convex, stochastic, robust)
Contribution Type: General: Most submissions will fall into this type.
Academic Integrity: I acknowledge that I have read the NeurIPS Handbook and commit to adhering to all policies in the Handbook (https://neurips.cc/Conferences/2026/MainTrackHandbook (https://neurips.cc/Conferences/2026/MainTrackHandbook)), the NeurIPS Code of Conduct and the NeurIPS Academic Integrity Policy.
LLM Usage: Editing (e.g., grammar, spelling, word choice)
Declaration: I confirm that the above information is accurate.
Reviewer Nomination: Davide Sartor (/profile?id=~Davide_Sartor1)
Submission Number: 23696

Discussion (?id=hBh55r5Asy#discussion)

Filter by reply type... Filter by author... Search keywords... Sort: Newest First

Everyone | Program Chairs | Submission23696... | Submission23696... | Submission23696... | Submission23696... | Submission23696... | Submission23696... | Submission23696... | ✖

25 / 25 replies shown

Add: **Withdrawal**

Paper Decision

Decision by Program Chairs 24 Sept 2026, 07:24 (modified: 24 Sept 2026, 12:26)

Program Chairs, Authors Revisions (/revisions?id=nahpl3LlPs)

Decision: Accept (poster)

Metareview:

This paper presents a functional Bayesian optimization method, referred to as 'L0MO' that represents each candidate as a sparse RKHS expansion and jointly optimizes the kernel coefficients and support locations, thereby searching over a richer nonlinear family than the fixed or expanding linear spans used by earlier FBO methods. The reviewers generally agree that the central idea is simple, intuitive, and potentially useful, that the reinterpretation of Vien et al.'s functional-gradient updates as natural-gradient descent in a fixed-support parameterization provides a valuable unifying perspective, and that the extensive experiments and functional benchmark library could benefit the community. The empirical results are OK, although the practical significance of the evaluation remains uncertain. The paper should clarify that the original BFO gradients use support locations from both the random initialization and previously evaluated functions, and it should present L0MO's novelty more narrowly as jointly selecting a sparse subspace through movable supports and an element within that subspace through its coefficients. Equation (3) appears to use the wrong EI sign for minimization, and Equation (4) contains a coefficient error. Moreover, the set of at-most-m-sparse expansions is not globally a smooth m(d+1)-dimensional manifold because of permutation symmetries, zero coefficients, coincident supports, and embedded lower-sparsity subsets. The motivation for fixing m is also unclear. No assumptions ensure existence of the acquisition maximizer, and no convergence, regret, or approximation guarantee explains when sparse optimization adequately represents the full-RKHS problem. Finally, the main results rely on hindsight selection of kernel profiles and RKHS lengthscales, while several ridge benchmarks are structurally aligned with sparse point-evaluation functionals. The author response did a good job, resolving most of reviewers' concerns. This leads that two of reviewers raise their overall score. Regarding the mainifold issue, the stratified set is more accurate, which is accepted by the authors. The very strong empirical evidence is overstated, which should be taken into account for the final version. In the references, "Sampling plans" [Alexander et al, 2008] is not a standalone book and should be Chapter 1 of the book 'Engineering Design via Surrogate Modelling: A Practical Guide' by Alexander I. J. Forrester, András Sóbester, and Andy J. Keane.

Final Justification:

The author response well resolved most of reviewers' concerns, leading that two of reviewers raise their overall score.

Citation Checker by Program Chairs

Citation Checker by Program Chairs 04 Aug 2026, 14:30 (modified: 05 Sept 2026, 01:17)

Program Chairs, Senior Area Chairs, Area Chairs, Authors Revisions (/revisions?id=firIehseQt)

Comment:

Automatically flagged references

Here we list the references that were automatically flagged by our checker. Note that our checker may miss many hallucinated references, and some of the listed references may have been flagged erroneously. **It is therefore crucial that you check them manually, following the guidance above. We rely on the Area Chairs to decide whether these constitute violations of responsible authorship.**

For papers falling into either of the following categories, we expect your default recommendation to be a rejection on the grounds of irresponsible authorship. This is strong presumption rather than a strict rule; as the Area Chair, you retain discretion, and may recommend otherwise where the specific circumstances of a paper warrant it:

1. The paper contains at least two hallucinated references according to the criteria described below the report.
2. The paper contains one hallucinated reference and there is some additional evidence of AI slop.

If a paper has only one problematic reference, but no additional concerns about the quality of the paper or its presentation, please simply alert the authors in your meta-review.

Please complete the form “Hallucinated References Investigation” to communicate the results of your investigation prior to completing your meta reviews.

Citation Checker Report

Reference Check Summary

1/42 problematic references.

List of Problematic References with Reasoning

```
@book{forrester2008sampling,
  author = {Alexander I. J. Forrester and András Sóbester and Andy J. Keane},
  title = {Sampling Plans},
  publisher = {John Wiley & Sons, Ltd},
  year = {2008}
}
```

The citation incorrectly presents 'Sampling Plans' as a standalone book title published by John Wiley & Sons in 2008. In reality, 'Sampling Plans' is Chapter 1 of the book 'Engineering Design via Surrogate Modelling: A Practical Guide' by Alexander I. J. Forrester, András Sóbester, and Andy J. Keane.

Hallucinated/problematic reference criteria

These are adopted from ICLR and ICML 2026 hallucinated reference guidance:

- If you are unable to find any document or publication with a title anywhere near the one provided in the reference, it should be considered hallucinated. Minor differences in title should not be considered evidence of a hallucinated reference.
- If the reference's authors are only slightly incorrect, e.g., a few subtle misspellings, missing a few authors, or wrong authorship order, we do not consider this to imply a hallucinated reference, but please let the submission's authors know in your meta-review that they should fix this issue.
- If the author list has additional issues, e.g., added authors that are not actually authors of that publication or missing a large fraction of authors, then the reference should be considered hallucinated.
- If the reference's venue is a real venue but is incorrect for that reference, we do not consider this to imply a hallucinated reference, but please tell the submission's authors in your meta-review, these include wrong journal name, arXiv ID, etc.
- If there is no evidence that the reference's venue exists, the reference should be considered hallucinated.
- Finally, we note that there are many generic references that refer to existing artifacts (e.g. "GPT-4o, OpenAI et al. 2024") but do not refer to any real document. These are fine.
- Even with the above guidelines in mind, there will likely be some borderline cases. Please bring those up with your SAC if you need additional guidance.

Official Review of Submission23696 by Reviewer La2M

Official Review by Reviewer La2M 22 Jun 2026, 06:10 (modified: 24 Sept 2026, 13:06)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer La2M

Revisions (/revisions?id=f6ApBEG21c)

Summary:
This paper proposes a method for Bayesian optimization in function spaces, i.e., optimization problems whose search space is a function space and the objective is a (nonlinear) functional of it. The search space is specified via subsets of a reproducing kernel Hilbert space (RKHS) with a bounded pseudo-norm, which is defined as a number of non-zero kernel basis elements. To solve the challenge of optimizing the resulting acquisition functionals, the paper proposes a natural gradient scheme which makes use of the RKHS geometry to determine updates on both the coefficients and the locations of the basis elements. Experiments demonstrate superior performance when compared with existing Bayesian functional optimization baselines.

Contribution Type: General: Most submissions will fall into this type.

Strengths And Weaknesses:

Strengths

From my understanding, one of main contributions of this paper with respect to the original Bayesian functional optimization (BFO) framework [39] is showing that freely optimizing a fixed (large enough) number of both coefficients α_i and locations of kernel sections $k(\cdot, x_i)$ forming the function-valued candidates $u = \sum_i \alpha_i k(\cdot, x_i)$ for BFO yields significant performance improvements than the original functional gradient formulation. In the original BFO, to optimize the acquisition functional (AF) by gradient descent (GD) over the RKHS at each BO round t, we start with a random input function $u_{t,0}$ and follow the AF gradient for a number of steps. It turns out that, due to the AF being itself a functional of the posterior mean and variance of a Gaussian process (GP) defined over the RKHS function space, the resulting functional gradients will themselves be finite linear combinations of the kernel sections of previously evaluated input functions in the dataset and the new kernel sections from the random initialization $u_{t,0}$ for GD. Therefore, GD will not change the locations of the random initialization nor add any new locations other than the ones forming the kernel sections in the previous evaluations. Only the coefficients (i.e., kernel weights) are changed. Therefore, reformulating the AF optimization problem as directly optimizing both coefficients and locations can lead to a better exploration of the solution space, as demonstrated by the significant performance gains in the experimental results.

In summary,

- the paper is mostly well written and organized;
- it highlights a significant limitation with previous Bayesian functional optimization frameworks;
- and experimental results show superior performance with respect to baselines on most benchmarks.

Weaknesses

- The presentation of the technical arguments of the paper can be a bit confusing at times. Sec. 3 seems to claim that GD on EI and UCB leads only to evolving coefficients of the “support functions” of the initial random input function of GD at a given BO round. These support functions can easily be interpreted as the kernel sections $k(\cdot, x)$ of only the random initialization, not the kernel sections of the previously evaluated input functions in the GP dataset. That would, however, be a misleading argument. It’s only in Sec. 4 around Eq. 15 that it’s clearly shown that the authors understand that the set of support functions includes those of both the initialization and the previously evaluated candidates. Although these might have been well explained in [39], I’d still suggest adding a couple sentences explaining the relationship between the kernel gradients and the AF gradients for the classic types of AF considered.
- In Sec. 4, it might not be immediately to the unfamiliar reader that the gradients of the kernels only depending on the current and the previously evaluated candidates leads to GD iterates that keep the set of kernel locations fixed. The notation in Eq. 15 is also quite ambiguous as to what x_i' and u' are referring to. Critically, the notation $\mathcal{D}$ is being abused, as it was previously referring to the domain of the candidate functions, not what appears to be the set of candidates in the dataset under my most plausible interpretation for Eq. 15.
- From a methodological point of view, the contribution of the paper might seem somewhat incremental with respect to BFO [39], as the search space is still a subset of an RKHS, with only the GD algorithm being changed. Hence, I’d suggest emphasizing in Sec. 4 what are the additional “degrees of freedom” that are gained by freely optimizing the locations in addition to the coefficients when compared to BFO. For instance, as far as I understand, BFO will, in principle, be searching over a finite-dimensional subspace of the RKHS of expanding dimensionality from round to round, while L0MO keeps the dimensionality of the space fixed. This may seem as a disadvantage. However, practically, this leads to a significant technical advantage, as it is equivalent to searching over both a set of finite-dimensional subspaces (by varying the kernel locations) and its respective elements (by varying the kernel coefficients). The number of parameters could be fixed a priori based on, e.g., precision requirements of the application, or alternatively be allowed to grow over time, by increasing m on a schedule across BO iterations (as suggested in Sec. 6), which might be able to guarantee some form of asymptotic convergence.
- As a smaller technical detail, the maximum of Eq. 5 might not exist in general, since the constrained RKHS subset is not necessarily compact, and even if it is compact (e.g., by taking coefficients from a compact interval and having $\mathcal{D}$ also compact), a general AF might not be continuous with respect to its topology. Establishing formal assumptions for the finiteness of such maxima might be subject of further theoretical work, but there’s no mention of these issues in the paper around the problem statement.
- Given its importance throughout the paper, I’d also suggest adding a more formal definition of the L^0 norm, which seems to be only briefly and informally defined in lines 114-118.
- Despite a few ablation studies in the appendix, there is little to no discussion of the experimental results in the main paper. For instance, there is no discussion on why Shilton et al.'s [35] method outperforms L0MO and the

baselines on the Brachistochrone problem.

Quality: 4: excellent
Clarity: 3: good
Significance: 4: excellent
Originality: 3: good
Questions:

- In lines, 117-118, it is implied that the constraint set can be a set in $\mathbb{R}^d$, but that doesn't seem correct, as it's a subspace of $\mathcal{H}$, and d is presented as the dimensionality of the input function's domain in the definition of k in line 112.
- What would be a more formal definition of the L^0 norm?
- In line 154, is n referring to the number of observations in the GP dataset or the number of kernel sections in the candidate functions (as in line 93)? The former makes more sense, but the previous use of n indicates the latter, which does not seem correct. I'd suggest revising the choice of notation. Maybe using upper and lower case, N and n, would improve it.

Limitations:
The main limitations are discussed in Sec. 6.

Rating: 5: Accept: Technically solid paper, with high potential value on at least one sub-area of AI or moderate-to-high impact on more than one area of AI, with good-to-excellent evaluation, resources, reproducibility, and no unaddressed ethical considerations.
Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
No noticeable formatting issues.

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes
Final Justification:
I have decided to increase my score after the rebuttal, given that my main concerns have been addressed. I'd, however, strongly suggest that the authors pay careful attention to implement the changes suggested in the discussions, especially the claims around the manifold construct, which might be misleading if not revised.

Official Rebuttal

Official Comment
by Authors (Meghan E. Huber (/profile?id=~Meghan_E._Huber1), Donghyun Kim (/profile?id=~Donghyun_Kim11), Nathan Wycoff (/profile?id=~Nathan_Wycoff2), Davide Sartor (/profile?id=~Davide_Sartor1))
28 Jul 2026, 16:59
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer La2M

Comment:
Thanks very much for your thorough review and we were very pleased that you saw the importance of the work.

Weaknesses

> The presentation of the technical arguments of the paper can be a bit confusing at times. [...] I'd still suggest adding a couple sentences explaining the relationship between the kernel gradients and the AF gradients for the classic types of AF considered.

Great point, that could easily be a source of confusion, and we will work on making this clearer.

> In Sec. 4, it might not be immediately [...] my most plausible interpretation for Eq. 15.

Great points; thanks for your attention to detail in surfacing these issues.

> From a methodological point of view, the contribution of the paper might seem somewhat incremental with respect to BFO [39], as the search space is still a subset of an RKHS, with only the GD algorithm being changed. Hence, I'd suggest emphasizing in Sec. 4 what are the additional "degrees of freedom" that are gained by freely optimizing the locations in addition to the coefficients when compared to BFO. For instance, as far as I understand, BFO will, in principle, be searching over a finite-dimensional subspace of the RKHS of expanding dimensionality from round to round, while L0MO keeps the dimensionality of the space fixed. This may seem as a disadvantage. However, practically, this leads to a significant technical advantage, as it is equivalent to searching over both a set of finite-dimensional subspaces (by varying the kernel locations) and its respective elements (by varying the kernel coefficients). The number of parameters could be fixed a priori based on, e.g., precision requirements of the application, or alternatively be allowed to grow over time, by increasing on a schedule across BO iterations (as suggested in Sec. 6), which might be able to guarantee some form of asymptotic convergence.

We think that the general idea given here is right, but just want to mention that

- we increase the dimension of the search manifold during the optimization algorithm linearly (details are in Appendix E)
- we have more direct control on the dimensionality of the search space compared to [39], we just choose to grow it at a similar rate to have a fair comparison.
- it's not just the GD algorithm that has changed, but also the constraint space: in BFO it was a unit ball of the RKHS intersected with a random subspace, in our work it is the L^0 manifold.
- in [39] a sparsification step is needed to make the approach feasible, intriducing a discrepancy between the optimization and evaluation.

> As a smaller technical detail, the maximum of Eq. 5 might not exist in general [...] there's no mention of these issues in the paper around the problem statement.

This is a great point and we'll add discussion of this issue to the article.

> Given its importance throughout the paper, I'd also suggest adding a more formal definition of the norm, which seems to be only briefly and informally defined in lines 114-118.

Good point; in response to your question below we provide mathematical formalism which we will fold into the camera ready version.

> Despite a few ablation studies in the appendix, there is little to no discussion of the experimental results in the main paper. For instance, there is no discussion on why Shilton et al.'s [35] method outperforms L0MO and the baselines on the Brachistochrone problem.

Thanks for this point, the results do warrant additional discussion; we'll use some of the camera-ready space to expand on the following main ideas. The brachistochrone problem is a challenging task, and the optimizing solution is trascendental and has no exact finite dimensional representation in the RKHS, making it not-obvious a priori that our method will find a useful sparse combination. The method of Shilton et al is tailored to perform well in 1d scenarios by using a dense grid to parametrize functions. Its candidates are also higher dimensional, which helps in this specific problem. Despite this, our method is a very close second.

Official Rebuttal

Official Comment

by Authors (Meghan E. Huber (/profile?id=~Meghan_E._Huber1), Donghyun Kim (/profile?id=~Donghyun_Kim11), Nathan Wycoff (/profile?id=~Nathan_Wycoff2), Davide Sartor (/profile?id=~Davide_Sartor1))

28 Jul 2026, 16:59 (modified: 29 Jul 2026, 16:19)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer La2M

Revisions (/revisions?id=h6cN8tECjY)

Comment:

Questions

> In lines, 117-118, it is implied that the constraint set can be a set in $\mathbb{R}^d$, but that doesn't seem correct, as it's a subspace of $\mathcal{H}$, and is presented as the dimensionality of the input function's domain in the definition of in line 112.

Thanks for pointing out that that passage was confusing. What we were trying to say is that our just-defined L^0 norm on an RKHS $\|\cdot\|_0^{\mathcal{H}}$ is qualitatively different than the canonical ℓ_0 norm on Euclidean d-space $\mathbb{R}^d$ denoted $\|\cdot\|_0^{\mathbf{R}^d}$ in that $\|\cdot\|_0^{\mathcal{H}}$ is a connected set, and so somewhat friendlier to local continuous optimization than is the infamously difficult to optimize $\mathbb{R}^d$ denoted $\|\cdot\|_0^{\mathbf{R}^d}$. It would indeed be clearer if we used a different letter than d which as you suggested is already used; we were trying to re-use an already defined object.

> What would be a more formal definition of the norm?

Definition (L^0 norm on an RKHS). Let $\mathcal{H}$ be the RKHS with kernel $k : \mathcal{D} \times \mathcal{D} \to \mathbb{R}$. For $u \in \mathcal{H}$, the L^0 norm of u is the smallest m such that u is exactly representable by m kernel sections:

$$\|u\|_0^{\mathcal{H}} = \inf\{m : \exists \alpha_i \in \mathbb{R}, x_i \in \mathcal{D}, \|u - \sum_{i=1}^{m} \alpha_i k(\cdot, x_i)\| = 0\}$$

That is, $|u|_0^{\mathcal{H}}$ is the smallest number of kernel sections needed to represent u exactly.

> In line 154, is referring to the number of observations in the GP dataset or the number of kernel sections in the candidate functions (as in line 93)? The former makes more sense, but the previous use of indicates the latter, which does not seem correct. I'd suggest revising the choice of notation. Maybe using upper and lower case, and , would improve it.

You're right that there is a conflict in notation; thanks very much for your detailed reading to bring this to our attention.

Thanks very much for your helpful feedback which has already improved our paper and we look forward to our continued discussion.

Replying to Official Rebuttal

Official Comment by Reviewer La2M

Official Comment by Reviewer La2M 02 Aug 2026, 10:41

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer La2M

Comment:

Thanks for the clarifications. My main concerns have been addressed. Under this definition for the L^0 norm and considering related concerns from other reviewers, isn't this norm just finite for members of the typical pre-RKHS formed by the linear span of the kernel, i.e., $H_0 = \left\{\sum_{i=1}^{m} \alpha_i k(\cdot, x_i) \mid \alpha_1, \ldots, \alpha_m \in \mathbb{R}, x_1, \ldots, x_m \in \mathcal{D}, m \in \mathbb{N}\right\}$? In this case, any L^0-constrained set is essentially a subset of H_0, which is a dense subset of $\mathcal{H}$. This is just an observation, however, and I'm not sure if it helps with the analysis. It could help, however, to connect readers with related concepts typically found in introductory material for RKHSs.

Replying to Official Comment by Reviewer La2M

Official Comment by Authors

Official Comment

by Authors (Meghan E. Huber (/profile?id=~Meghan_E._Huber1), Donghyun Kim (/profile?id=~Donghyun_Kim11), Nathan Wycoff (/profile?id=~Nathan_Wycoff2), Davide Sartor (/profile?id=~Davide_Sartor1))

03 Aug 2026, 22:32

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer La2M

Comment:
Yes, your observation is exactly right: the limiting set of our searchspace $\lim_{m\to\infty} \bigcup_m \mathcal{M}^m$ (i.e. all elements with finite L_0 norm) is exactly the pre-RKHS $\mathcal{H}_0$.

The fact that $\mathcal{H}_0$ is dense in $\mathcal{H}$, is at the core of the asymptotical approximation guarantees argument we sketched in response to reviewer Zbhf.

We totally agree that we should have stated this explicitly, as it might not be immediatly obvious to readers unfamiliar with RKHS. In general we will use the extra camera ready space to provide a bit more background on RKHS concepts.

Thanks again for this review process that has very helpfull in improving the quality the paper.

Official Review of Submission23696 by Reviewer Zbhf

Official Review by Reviewer Zbhf 19 Jun 2026, 18:57 (modified: 24 Sept 2026, 13:06)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Zbhf

Revisions (/revisions?id=oUh9Hexo1D)

Summary:
This paper studies Bayesian optimization in functional spaces through a sparse functional representation. It optimizes expected improvement over functions in a RKHS with finite number of support points. The method jointly optimizes both the parameters associated with the function representation and the locations of the support points, yielding a flexible functional search procedure.

The paper provides a principled derivation of the method, discusses its geometric interpretation, and develops optimization procedures in the resulting functional space. Empirically, the approach is evaluated on a collection of synthetic and application-inspired benchmark tasks, where it consistently outperforms several competing methods in terms of optimization performance and regret.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:
The paper is a welcomed submission on using RKHS spaces to perform Functional Bayesian Optimization (FBO). I found the methodology generally sound, and the paper does a good job explaining how Bayesian optimization concepts can be extended to functional spaces. In particular, the optimization procedure over support points and function parameters is intuitive and technically well developed. Useful properties of RKHS spaces and natural gradients are adequately explained.

A notable strength is the flexibility obtained by jointly optimizing both the support-point locations and the associated function representation. This provides a richer search space than fixed discretizations and appears to contribute to the empirical gains observed in the experiments. Reformulating existing litterature as a special case of this method is interesting and valuable. The empirical results are generally strong, with consistent improvements over competing baselines across the reported benchmarks.

My main concerns relate to the modeling assumptions and the experimental evaluation.

First, the method enforces a fixed sparsity level by restricting the representation to exactly (M) support points. While this is computationally convenient, it is less clear why this constraint is desirable from an optimization perspective. If the objective is to optimize over functions, one might expect the representation complexity to grow as more information is acquired. This raises the question of whether support points should be dynamically added or removed during optimization rather than fixed a priori. The ability to move support points is compelling, but it seems equally natural to allow the complexity of the representation itself to adapt over time.

The authors acknowledge this direction as future work, which makes me wonder whether the (ℓ_0) constraint should play such a central role in the formulation of the method. If the constraint is primarily introduced for computational tractability, rather than because sparsity is intrinsically desirable, then the ideal optimization problem would arguably still be posed over the full RKHS, with the sparse representation viewed as an approximation. In that case, I believe the framing of the method should place less emphasis on the (ℓ_0)-constrained formulation itself.

Similarly, while it is technically correct that the (ℓ_0) constraint induces a manifold structure, I found this aspect somewhat overstated. The manifold viewpoint does not seem to play a substantial role in either the development of the method or the intuition behind it, and I am not convinced that it adds much understanding beyond the sparse parameterization itself.

My largest concern is the significance of the empirical evaluation. Although the benchmarks are diverse, many of them remain relatively synthetic or toy problems. Examples such as optimizing activation functions for MNIST models or low-dimensional control tasks provide useful testbeds, but they do not fully demonstrate that the method solves practically important functional optimization problems. Consequently, it is difficult to assess whether the observed gains would persist in realistic settings where evaluations are noisy, expensive, and structurally complex.

Relatedly, I found it somewhat concerning that performance on the more challenging settings appears substantially weaker than on the synthetic benchmarks. Additional experiments on realistic noisy optimization problems would strengthen the paper considerably and help establish the practical relevance of the method.

Overall, I found the paper technically solid and clearly written, with promising empirical results. However, I remain unconvinced about the practical significance of the current evaluation and would like stronger evidence that the method provides meaningful advantages on realistic functional optimization problems.

Quality: 3: good
Clarity: 4: excellent
Significance: 3: good
Originality: 3: good
Questions:

1. The method fixes the number of support points (M) throughout optimization. Beyond computational considerations, what is the conceptual justification for enforcing a fixed sparsity level? Would allowing adaptive addition or removal of support points improve optimization performance?
2. How sensitive is performance to the choice of (M)?
3. Many of the benchmarks are synthetic or application-inspired rather than truly real-world optimization tasks. Can the authors provide additional evidence that the method remains effective on realistic noisy functional optimization problems?
4. What is the computational cost of the method? An answer on size of data, hardware and runtime would be useful here

A convincing response addressing Questions 1 and 3 would substantially increase my confidence in the paper. In particular, evidence that the fixed sparsity constraint is not a limiting modeling assumption and that the method performs well on realistic noisy tasks would strengthen the significance assessment.

Limitations:
yes

Rating: 5: Accept: Technically solid paper, with high potential value on at least one sub-area of AI or moderate-to-high impact on more than one area of AI, with good-to-excellent evaluation, resources, reproducibility, and no unaddressed ethical considerations.
Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
N/A

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes
Final Justification:
The authors have addressed most of my concerns, including straightening the evaluation and the manifold framing. I have upgraded my rating accordingly

Official Rebuttal

Official Comment

by Authors (Meghan E. Huber (/profile?id=~Meghan_E._Huber1), Donghyun Kim (/profile?id=~Donghyun_Kim11), Nathan Wycoff (/profile?id=~Nathan_Wycoff2), Davide Sartor (/profile?id=~Davide_Sartor1))

28 Jul 2026, 17:03 (modified: 28 Jul 2026, 17:08)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Zbhf

Revisions (/revisions?id=dv6VXudVk1)

Comment:

Thanks very much for taking the time to carefully read our article and for providing us with your feedback; we are very grateful. We will below comment on your weakness and questions feedback.

Weaknesses

> First, the method enforces a fixed sparsity level by restricting the representation to exactly (M) support points. [...] I believe the framing of the method should place less emphasis on the ()-constrained formulation itself.

We could not agree more, and in fact we do grow the number of support points used, from 1 to 10 at a linear rate, throughout the course of the optimization. Our formulation of working with nested submanifolds allows for these runs from different dimensional spaces to be seamlessly folded together (See Appendix E for more information). An ideal schedule would also adapt dynamically to the observations, rather than the fixed linear growth we currently use; this is a research direction we are currently actively pursuing.

> Similarly, while it is technically correct that the constraint induces a manifold structure, I found this aspect somewhat overstated. The manifold viewpoint does not seem to play a substantial role in either the development of the method or the intuition behind it, and I am not convinced that it adds much understanding beyond the sparse parameterization itself.

The primary reason to mention the manifold structure, which in and of itself is not a choice but merely inherent to our method, is to emphasize the contrast to the linear subspaces used by all state of the art BFO methods, but also to provide a novel interpretation of [39] among other uses.

> My largest concern is the significance of the empirical evaluation. Although the benchmarks are diverse, many of them remain relatively synthetic or toy problems. Examples such as optimizing activation functions for MNIST models or low-dimensional control tasks provide useful testbeds, but they do not fully demonstrate that the method solves practically important functional optimization problems. Consequently, it is difficult to assess whether the observed gains would persist in realistic settings where evaluations are noisy, expensive, and structurally complex.

In the functional setting, there is not yet a library of established problems, and we worked hard to produce 4 realistic test benchmarks, whereas previous work [39; Vien et al] and [38, Vellanki et al], [35; Shilton et al] had only 2 realistic test benchmarks. We we have double the realistic test functions than previous works we are comparing aganist. In addition, though previous work did not do this reliably, we have all of our test functions easily available from our code, and we hope the realistic benchmarks used in this paper can become a standard for future functional BO.

Secondly, the realistic problems we use are not toy problems. The pinhole problem is a significant challenge in computational engineering which was suggested on the basis of subject matter expertise. The MNIST dataset is a classical ML dataset, but we are not solving the MNIST regression merely via backprop, we are optimizing the activation function of a neural network as a black box, few shots optimization. This is not at all a trivial task. Similarly, the optimal brachistochrone is a trascendental problem, and we are trying to optimize the curve only using sparse information on the travel time.

Nevertheless, we agree that it is always better if possible to provide more experiments, and therefore have worked hard to rapidly provide, in direct response to this concern, a fifth realistic benchmark consisting in a robotic control task. It is a planar single-leg hopping robot with rigid ground contact and Coulomb friction: the optimizer searches over the vector valued reference trajectory for the hip and knee joints, which is tracked by a joint space PD controller with torque limits, and the objective is the negative apex height reached after the robot has landed at least once. This is a genuinely hard functional objective, with discontinuity in dynamics while making contact with the ground vs airborne.

We provide results below (lower is better, median with 95% confidence interval over seeds):

Method	Final regret	Average regret
Random	-1.869 [-1.957, -1.810]	-1.637 [-1.700, -1.614]
Shilton	-1.712 [-1.737, -1.673]	-1.621 [-1.648, -1.589]
Kundu	-1.613 [-1.735, -1.475]	-1.604 [-1.729, -1.445]

Method	Final regret	Average regret
Vien	-1.499 [-1.505, -1.482]	-1.455 [-1.462, -1.440]
Vellanki	-1.467 [-1.467, -1.467]	-1.458 [-1.460, -1.455]
Ours	**-3.092 [-3.164, -1.749]**	**-1.801 [-2.300, -1.655]**

Our method is the clear winner on this benchmark, and it is the only method that finds trajectories which get the robot substantially off the ground; the baselines all cluster near the height reached by simply not falling over.

Official Rebuttal

Official Comment

by Authors (Meghan E. Huber (/profile?id=~Meghan_E._Huber1), Donghyun Kim (/profile?id=~Donghyun_Kim11), Nathan Wycoff (/profile?id=~Nathan_Wycoff2), Davide Sartor (/profile?id=~Davide_Sartor1))

28 Jul 2026, 17:11 (modified: 28 Jul 2026, 17:21)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Zbhf

Revisions (/revisions?id=atinwOs5KX)

Comment:

> Relatedly, I found it somewhat concerning that performance on the more challenging settings appears substantially weaker than on the synthetic benchmarks. Additional experiments on realistic noisy optimization problems would strengthen the paper considerably and help establish the practical relevance of the method.

We do not agree that our method performs poorly on the real test problems: our method comes first in 3/4 problems and second on that 4th problem, where it is a very close second. The wins from our method on Pendulum and Pinwheel are substantial, and while the final value of our MNIST win is close to some other methods', the early trajectory on MNIST favors our method clearly. We hope that the addtion of a fifth benchmark alleviates any remaining concern.

> Overall, I found the paper technically solid and clearly written, with promising empirical results. However, I remain unconvinced about the practical significance of the current evaluation and would like stronger evidence that the method provides meaningful advantages on realistic functional optimization problems.

We hope that we were able to alleviate some of your concerns, and thanks very much for your careful review which has already improved our article.

Questions

> The method fixes the number of support points (M) throughout optimization. Beyond computational considerations, what is the conceptual justification for enforcing a fixed sparsity level? Would allowing adaptive addition or removal of support points improve optimization performance?

First we'd like to again clarify that we in fact have an increasing schedule for M. In general, the dimension of the approximating space is difficult to set and presents a challenge for all methods, including ours. Consequently, determining a robust strategy for choosing M would benefit all methods and would be an important but distinct contribution from our article.

> How sensitive is performance to the choice of (M)?

The performance of all Functional BO methods, including ours, is contingent on selecting M reasonably. We view dimension selection as a major outstanding problem for FBO. We agree that investigating the effect of the schedule is an important point we have insufficiently studied. To fix this, we have now run an ablation study comparing the growing schedule with

fixed schedules of M=1, M=5 and M=10. For space constraints, here we report only the results on the realistic benchmarks, but we have run the same ablation also for the remaining benchmarks (lower is better, median with 95% confidence interval over seeds):

	Brachistochrone [10^{-1}]		**Pendulum [10^{2}]**		**Hopper [10^{0}]**	
schedule	Average Regret	Final Regret	Average Regret	Final Regret	Average Regret	Final Regret
Fixed M=1	1.86 [1.82, 1.88]	1.71 [1.71, 1.71]	5.32 [5.30, 5.33]	5.27 [5.27, 5.27]	-1.45 [-1.45, -1.44]	-1.47 [-1.47, -1.47]
Fixed M=5	1.06 [0.94, 1.51]	0.38 [0.31, 0.68]	4.83 [4.77, 4.87]	4.55 [4.52, 4.61]	**-2.33 [-2.40, -2.15]**	**-2.65 [-2.72, -2.54]**
Fixed M=10	3.01 [2.84, 3.16]	2.42 [2.24, 2.73]	**4.67 [4.55, 4.79]**	**4.28 [4.14, 4.41]**	-2.06 [-2.35, -2.02]	-2.53 [-2.76, -2.25]
Growing (ours)	**0.82 [0.77, 0.86]**	**0.30 [0.25, 0.31]**	5.06 [4.95, 5.13]	4.51 [4.38, 4.67]	-1.83 [-2.11, -1.62]	-2.28 [-2.68, -1.80]

As expected, M=1 saturates very early on every benchmark, confirming that a rich enough representation is necessary. The larger fixed schedules sometimes outperform the growing one, but neither is uniformly better. Similar results are true for the other benchmarks. The growing schedule is seldom the optimal, but is never far from the best on any benchmark enabling sample-size robustness: it behaves like a small M while data is scarce and like a large M once there is enough data to support it.

That said, these results also show that no single schedule dominates, which is supports our shared view that adaptive dimension selection is an important problem. MNIST and PinWheel are too computationally intensive and we were unable to collect all results without delaying this discussion.

> Many of the benchmarks are synthetic or application-inspired rather than truly real-world optimization tasks. Can the authors provide additional evidence that the method remains effective on realistic noisy functional optimization problems?

As we have mentioned in our view we have provided significant evidence already that this method can perform well on real noisy functional optimization problems. We also provide an additional benchmark.

Replying to Official Rebuttal

Official rebuttal

Official Comment

by Authors (Meghan E. Huber (/profile?id=~Meghan_E._Huber1), Donghyun Kim (/profile?id=~Donghyun_Kim11), Nathan Wycoff (/profile?id=~Nathan_Wycoff2), Davide Sartor (/profile?id=~Davide_Sartor1))

28 Jul 2026, 17:11

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Zbhf

Comment:

> What is the computational cost of the method? An answer on size of data, hardware and runtime would be useful here

We ran this study on a heterogeneous computing cluster, which complicates timing studies because of potential hardware differences across method/problem combinations, but this was unavoidable given that we had so many different hyperparameter settings to test. For this reason, we did not conduct a formal timing study of our main experiments.

However, you make a great point that this should be studied more carefully, and we have thus performed a timing experiment, fixing the hardware. All runs were on single core CPU.The total wall clock time per run, in seconds, averaged across all benchmarks, is:

method	Surrogate fit	Acquisition
Random	<0.01	3.9
Kundu	85.5	254.9
Vien	88.5	586.1
Shilton	81.1	891.7
Vellanki	154.1	79.7
Ours	**86.9**	**202.2**

Our method is squarely in the middle of the pack: consistently cheaper than Vien, cheaper than Kundu, and more expensive than the grid based methods, which buy their speed by fixing the support locations and so give up our model's additional flexibility. We should also mention that these numbers represent the average across a run with fixed total acquisition budget, and that our method actually scales better than the competitors as the total budget increases, meaning that this timing study we have executed was by design unfavorable to our method.

In the updated manuscript, we will add an additional section in the appendix with in depth breakdown of wall clock times across different benchmarks, as there are slight variations in the acquisition time.

--

We want to thank you again for sharing your perspective which has already improved our paper, and we look forward to the opportunity to discuss more during this upcoming review period.

Replying to Official rebuttal

Official Comment by Reviewer Zbhf

Official Comment by Reviewer Zbhf 29 Jul 2026, 17:04

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Zbhf

Comment:

Thank you for the detailed rebuttal. I found it very responsive to my concerns.

To clarify my original concern regarding the experiments, it was not only whether the optimized functional form was difficult to recover, but also whether recovering it was practically significant. I agree that black-box optimization of a neural-network activation function is technically challenging; my uncertainty was primarily whether this constitutes an important practical functional optimization problem. Regardless, the additional experiments, together with the ablations and runtime results, provide convincing empirical evidence, and I now consider my concerns regarding the evaluation and the choice of (M) to be largely addressed.

My remaining concern is the manifold framing. I encourage the authors to directly address the area chair's concerns and to reconsider how central the manifold viewpoint should be to the paper. The fact that the search space is nonlinear rather than a linear subspace seems more like a useful observation than a defining feature of the method, while the technical issue raised by the area chair is substantial.

A convincing clarification of this point would raise my assessment to accept.

Official Comment by Authors

Official Comment

by Authors (Meghan E. Huber (/profile?id=~Meghan_E._Huber1), Donghyun Kim (/profile?id=~Donghyun_Kim11), Nathan Wycoff (/profile?id=~Nathan_Wycoff2), Davide Sartor (/profile?id=~Davide_Sartor1))

31 Jul 2026, 18:23 (modified: 31 Jul 2026, 18:32)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Zbhf

Revisions (/revisions?id=g1903aikwY)

Comment:

> Thank you for the detailed rebuttal. I found it very responsive to my concerns.

We are likewise very grateful for your engagement which has already lead to a much improved paper.

> My remaining concern is the manifold framing. I encourage the authors to directly address the area chair's concerns and to reconsider how central the manifold viewpoint should be to the paper. The fact that the search space is nonlinear rather than a linear subspace seems more like a useful observation than a defining feature of the method, while the technical issue raised by the area chair is substantial.

We believe the manifold viewpoint helps us build a clear distinction between our work and prior literature, and it was also very helpful to us in developing and implementing this methodology, but we respect and understand your alternative perspective. In our opinion, describing our search space as an uncountable collection of linear subspaces would only apply to the canonical parametrization, and it is not this perspective which led to our algorithm. Conversely, the manifold viewpoint allow us to reframe [39] as a special case of our method, obtained by using a very specific and overly rigid parametrization. This viewpoint also higlights that our search space is not closed under addition which is a significant aspect to highlight given what the FBO community is used to.

We are very grateful to this review process for pointing out that more detail was needed regarding the technical issues summarized in the metareview, and we are very pleased to present a more precise descriptionthe of the elementary manifold-theoretic structure of our method, which we now list in response to Metareview quotations.

> METAREVIEW: the set of at-most-m-sparse expansions is not globally a smooth m(d+1)-dimensional manifold because of permutation symmetries, zero coefficients, coincident supports, and embedded lower-sparsity subsets.

The statement in 172-178 is imprecise as presented. As we expounded as a response to prod from reviewer mDJr to look more deeply into this section, the dimension of $\mathcal{M}^m$ is not locally constant, making it actually a union of manifolds.

To be more precise, $\mathcal{M}^m$ is a stratified set, composed of the L^0 level sets

$$\mathcal{S}^m = \{u \in \mathcal{H} : |u|_0 = m\}.$$

Each $\mathcal{S}^m$ is indeed a smooth manifold (for the rbf kernel we are considering). The canonical parameterization $\varphi(w) = \sum_{i=1}^m \alpha_i k(\cdot, x_i)$, $w = (\alpha, x)$, is an $m!$-sheeted covering mapping the open set $W^m = \{x_i \text{ pairwise distinct}, \ \alpha_i \neq 0\} \subset \mathbb{R}^{(d+1)m}$ onto $\mathcal{S}^m$ m!-to-1

We would like to stress that this does not affect our analysis in Section 4, which remains valid.

We also think it's a great idea to make explicit the details regarding the existence of the acquisition optimum subsequently mentioned by the Metareview.

Regarding convergence of the approximate problems, the family $\{\mathcal{M}^m\}_{m\in\mathbb{N}}$ is dense in H, and this is what allows us to conclude that the optimal solution in $\mathcal{M}^m$ converges in to the optimal solution in H as m increases.

Regarding existence of an acquisition maximizer, it is true that the maximizer in Eq.5 might not exist in some pathological cases where the acquisition candidate wanders to infinity. However,

1. the search space can typically be restricted to a compact set anyway (as we describe in Appendix E and note in line 20 for BO) and the acquisition function $\mathcal{I}$ is typically at least continuous (and also lipschitz under mild conditions). This is sufficient to guarantee the problem is well posed.
2. Even without this, the maximum for EI is guaranteed to exists under more general mild conditions.

We will use the additional space in the camera-ready version to add some discussion of this.

> A convincing clarification of this point would raise my assessment to accept.

We hope that our replies have clarified any missing concerns.

Replying to Official Comment by Authors

Official Comment by Reviewer Zbhf

Official Comment by Reviewer Zbhf 31 Jul 2026, 19:02

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Zbhf

Comment:

Thank you for your response, I will not argue further on the manifold framing I see your point.

I will defer to mDJr for the technical aspect of the manifold as I cannot judge that myself. I will upgrade to accept once confirmed.

Replying to Official Comment by Reviewer Zbhf

Official Comment by Authors

Official Comment

by Authors (Meghan E. Huber (/profile?id=~Meghan_E._Huber1), Donghyun Kim (/profile?id=~Donghyun_Kim11), Nathan Wycoff (/profile?id=~Nathan_Wycoff2), Davide Sartor (/profile?id=~Davide_Sartor1))

03 Aug 2026, 22:32

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Zbhf

Comment:

Thanks for your engagement and for the constructive discussion.

For your reference, Reviewer mDJr has now confirmed that the geometric expression has been corrected in the appropriate direction.

We are glad we were able to address your concerns, and we are grateful for your review, which prompted several of the additions in the metareview summary above.

Official Review of Submission23696 by Reviewer mDJr

Official Review by Reviewer mDJr 10 Jun 2026, 01:58 (modified: 24 Sept 2026, 13:06)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer mDJr

Revisions (/revisions?id=CQeEHyQjPG)

Summary:

This paper studies Functional Bayesian Optimization (FBO), where the optimization variable is a function rather than a finite-dimensional vector. The authors propose L0 Manifold Optimization (L0MO), which searches over sparse subsets of an RKHS by representing candidate functions as finite kernel expansions with jointly optimized coefficients and support locations. The paper also provides a unifying interpretation of prior work, especially

interpreting Vien et al. as natural gradient descent in a fixed-support parameterization, and introduces functional ridge benchmarks derived from standard finite-dimensional BO test functions. The experimental evaluation compares L0MO against several FBO baselines on projection, functional ridge, and more realistic tasks.

Contribution Type: General: Most submissions will fall into this type.
Strengths And Weaknesses:
Strengths:

- The main idea is clear and potentially useful. The paper identifies an existing limitation of prior FBO methods: many of them effectively optimize within a fixed finite-dimensional span, even when motivated as function-space optimization. The proposed method instead jointly optimizes support locations and coefficients in a sparse RKHS expansion, which is a simple and natural way to obtain a richer candidate class.
- The explanation of existing FBO work is novel. The paper gives a useful interpretation of Vien et al.'s functional gradient updates r[39] as natural gradient descent in a finite-dimensional fixed-support parameterization. Even though this does not become the main practical advantage of L0MO, it helps organize prior methods under a common geometric view.
- Experimental coverage is broad. The empirical section includes simple projection tasks, functional ridge versions of standard BO benchmarks, and several more applied tasks, including brachistochrone, MNIST activation optimization, pendulum control, and pinwheel trajectory optimization.
- Benchmark contribution may be useful to the community. The functional ridge construction provides a systematic way to turn familiar finite-dimensional BO benchmarks into FBO problems, and the authors state that they release an implementation of more than 70 benchmark functions and their functional extensions. This could become a useful testbed for future FBO research.

Weaknesses:

- The expected improvement formula in Eq. (3) appears to use the wrong sign for minimization. The paper defines $\gamma(u) = (\mu_{\mathrm{GP}}(u) - y^*)/\sigma_{\mathrm{GP}}(u)$, whereas standard minimization EI uses $(y^* - \mu_{\mathrm{GP}}(u))/\sigma_{\mathrm{GP}}(u)$. As written, the formula would not reward candidates with posterior mean below the current best value. In addition, the RKHS norm expansion in Eq. (4) seems to contain a coefficient typo in the first term: it appears to use $\alpha_i \alpha_j' k(x_i, x_j)$, whereas the first self-inner-product term should involve $\alpha_i \alpha_j k(x_i, x_j)$.
- The "manifold" claim may be overstated. The paper states that $\mathcal{M}^m = u \in \mathcal{H} : |u|_0 \leq m$ is an $(d+1)m$-dimensional manifold. This needs more qualification. The parameterization has permutation symmetries, possible coincident support points, zero coefficients, and lower-sparsity strata embedded in higher-sparsity sets, thus $\mathcal{M}^m$ is not a smooth manifold globally. The current statement around lines 172–178 is too strong.
- The paper is mainly methodological and empirical. It does not provide regret bounds, convergence guarantees, sample-complexity analysis, or approximation guarantees showing when optimizing over $\mathcal{M}^m$ is sufficient.
- The main experiments use hindsight hyperparameter selection. The appendix states that the surrogate kernel profile is selected per method and benchmark according to lowest final regret, and that the RKHS lengthscale is selected according to the best target value achieved across runs and methods. This would not be available in a real black-box optimization setting. A fixed default configuration, or sensitivity/ranking over all hyperparameter settings would make the conclusions more convincing.
- Benchmark and comparison fairness require more discussion. The functional ridge benchmarks are constructed using sparse RKHS linear functionals, reducing to evaluations $u(x_{ij})$ via the reproducing property. This structure seems particularly aligned with L0MO's inductive bias of movable sparse kernel supports. The authors would be better off including more "less sparse-RKHS-friendly" tasks or explicitly specifying which structural assumptions these benchmarks measure.

References:

r[39] Ngo Anh Vien, Heiko Zimmermann, and Marc Toussaint. Bayesian functional optimization. AAAI, 2018.

Quality: 3: good
Clarity: 3: good
Significance: 3: good
Originality: 3: good
Questions:
Given weakness 2, under what assumptions is $\mathcal{M}^m$ a smooth $(d+1)m$-dimensional manifold? How are permutation invariance, coincident support locations, zero coefficients, and lower-sparsity representations handled?

Limitations:

The authors discussed some limitations in the manuscript, but some other issues raised in the **weaknesses** section require further discussion.

Rating: 4: Borderline accept: Technically solid paper where reasons to accept outweigh reasons to reject, e.g., limited evaluation. Please use sparingly.
Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
Ethical Concerns: NO or VERY MINOR ethics concerns only
Paper Formatting Concerns:
NA

Code Of Conduct Acknowledgement: Yes
Responsible Reviewing Acknowledgement: Yes
Final Justification:
The supplementary experimental results effectively addressed my concerns about the fairness of the experiment. I think the paper meets the acceptance criteria, but the theoretical expression still needs further refinement, and the hyperparameter settings require quantitative analysis. If the authors address these two issues in subsequent stages, I will consider upgrading it to clear acceptance based on the theoretical completeness and the hyperparameter sensitivity analysis results.

Official Rebuttal

Official Comment
by Authors (Meghan E. Huber (/profile?id=~Meghan_E._Huber1), Donghyun Kim (/profile?id=~Donghyun_Kim11), Nathan Wycoff (/profile?id=~Nathan_Wycoff2), Davide Sartor (/profile?id=~Davide_Sartor1))
28 Jul 2026, 17:16
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer mDJr

Comment:
Thanks for your careful reading of our article; we have already greatly benefitted from your efforts. Below please find our responses to the weaknesses and questions.

> The expected improvement formula in Eq. (3) appears to use the wrong sign for minimization. The paper defines , whereas standard minimization EI uses . As written, the formula would not reward candidates with posterior mean below the current best value. In addition, the RKHS norm expansion in Eq. (4) seems to contain a coefficient typo in the first term: it appears to use , whereas the first self-inner-product term should involve .

Thanks so much for your careful reading for catching this; we will fix it.

> The “manifold” claim may be overstated. The paper states that is an -dimensional manifold. This needs more qualification. The parameterization has permutation symmetries, possible coincident support points, zero coefficients, and lower-sparsity strata embedded in higher-sparsity sets, thus is not a smooth manifold globally. The current statement around lines 172–178 is too strong.

Thanks for pointing this out; we debated discussing these issues but ultimately wanted to maximize the readability of our article. However, you have made us realize it would indeed be helpful to break down the basic manifold-theoretic properties of the set $\mathcal{M}^k$. We will add discussion of the elementary manifold-theoretic properties of the approximation space to the camera ready version by expanding on our treatment of this in our below answer to your question.

> The paper is mainly methodological and empirical. It does not provide regret bounds, convergence guarantees, sample-complexity analysis, or approximation guarantees showing when optimizing over is sufficient.

Indeed this article is mainly a methods contribution; FBO in general is far from having the same level of theoretical understanding as classical BO, and any significant advance in this area would very much merit its own dedicated article.

> The main experiments use hindsight hyperparameter selection. The appendix states that the surrogate kernel profile is selected per method and benchmark according to lowest final regret, and that the RKHS lengthscale is selected according to the best target value achieved across runs and methods. This would not be available in a real black-box optimization setting. A fixed default configuration, or sensitivity/ranking over all hyperparameter settings would make the conclusions more convincing.

This is a problem pervasive in functional BO, all of which have significant hyperparameter settings. Our study has elevated this previously underfocused dimension to the forefront because we agree it is critical to get it right, which is why we conducted such a thorough hyperparameter sweep. We chose to look at the setting which produced the best solution we have found across all methods because we want to understand which method has the best potential, and therefore deserves the most attention for developing automatic hyperparameter setting methods in the future.

Finally, every alternative has issues, including the two suggested. Picking a default configuration as you mention, and which was the default prior to this work, would not survive contact with our wildly different test functions and would inevitably favor certain methods. We initially had the same instinct to do hyperparameter sensitivity over many different hyperparameter settings, but we found that many of these settings achieved almost no optimization gain, and large differences in hopelessly miscalibrted methods dominated the sensitivity analyses and where not informative about which method is actually most performant.

This being said, regardless of how the approximation lengthscale is chosen, the relative rankings remain mostly unaffected (some of the close ties get broken, but our method is the top performer in most benchmarks).

Official Comment by Authors

Official Comment

by Authors (Meghan E. Huber (/profile?id=~Meghan_E._Huber1), Donghyun Kim (/profile?id=~Donghyun_Kim11), Nathan Wycoff (/profile?id=~Nathan_Wycoff2), Davide Sartor (/profile?id=~Davide_Sartor1))

28 Jul 2026, 17:20

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer mDJr

Comment:

> Benchmark and comparison fairness require more discussion. The functional ridge benchmarks are constructed using sparse RKHS linear functionals, reducing to evaluations via the reproducing property. This structure seems particularly aligned with L0MO's inductive bias of movable sparse kernel supports. The authors would be better off including more "less sparse-RKHS-friendly" tasks or explicitly specifying which structural assumptions these benchmarks measure.

Thanks for raising this point; we wish first to point out that it applies only to the ridge benchmarks, so this does not affect our realistic application functions that our method also performs well on, and which we think carry the most weight.

We do nevertheless agree it would be very interesting to test other types of ridge functions that are not sparse RKHS functionals, and are in the process of refactoring to including this in our next round of results. Concretely, we consider linear operators consisting of integrals against a given weighting functions w_i,

$$g_i = \int f(x), w_i(x), dx = \mathbb{E}_{x\sim w_i}\big[f(x)\big]$$

This can be seen as an extension of the sparse evaluation we were doing, which can be seen as monte carlo approximation of the integral as

$$g_i ; \approx ; \frac{1}{n}\sum_{j=1}^{n} f(x_j), \qquad x_j \sim w_i,$$

with more samples giving a better estimate, with the variance vanishing at the usual $1/n$ rate. If we choose w_i to be a random combination of trigonometric polynomials, the integral can instead be computed in closed form in Fourier space. We have updated the ridge benchmarks to use this non-sparse evaluation (i.e. linear functional that is not expressible as a sparse combination of evaluations of the original function).

Results on the four ridge benchmarks are below (median over seeds, with the 95% CI on the median; lower is better):

	Gramacylee [10^{-1}]		**Ackley [10^{1}]**		**Hartman [10^{0}]**		**Rosenbrock [10^{4}]**	
method	Average Regret	Final Regret	Average Regret	Final Regret	Average Regret	Final Regret	Average Regret	Final Regret
Random	5.53 [5.38, 5.61]	5.19 [5.14, 5.24]	1.58 [1.55, 1.59]	1.46 [1.43, 1.51]	3.14 [3.11, 3.16]	3.04 [3.00, 3.07]	6.57 [6.51, 6.63]	6.10 [5.92, 6.21]
Kundu	6.50 [6.48, 6.66]	6.37 [6.37, 6.37]	1.60 [1.58, 1.61]	1.51 [1.49, 1.53]	2.93 [2.90, 2.95]	2.65 [2.62, 2.68]	5.89 [5.83, 5.93]	5.33 [5.28, 5.37]
Vien	5.45 [5.39, 5.53]	5.13 [5.13, 5.13]	1.60 [1.56, 1.64]	1.54 [1.39, 1.59]	2.53 [2.51, 2.57]	2.37 [2.30, 2.40]	5.39 [5.33, 5.49]	4.75 [4.68, 4.89]
Shilton	5.50 [5.38, 5.63]	5.13 [5.13, 5.13]	1.39 [1.35, 1.41]	1.15 [1.07, 1.22]	2.58 [2.52, 2.59]	2.42 [2.38, 2.46]	5.05 [4.97, 5.13]	4.42 [4.30, 4.46]
Ours	**4.35 [4.01, 5.04]**	**3.42 [2.08, 4.35]**	**0.66 [0.61, 0.84]**	**0.09 [0.06, 0.25]**	**1.36 [1.35, 1.37]**	**0.78 [0.78, 0.78]**	**3.04 [2.90, 3.20]**	**0.20 [0.19, 0.22]**

Our method outperforms the baselines on this revised version of the benchmark as well, on both average and final regret and across all four ridge targets.

> Given weakness 2, under what assumptions is a smooth -dimensional manifold? How are permutation invariance, coincident support locations, zero coefficients, and lower-sparsity representations handled?

Thanks for bringing up these great questions. The mapping δ from a particular set of coefficients $\alpha_1, \ldots, \alpha_m$ and locations $x_1, \ldots, x_m$ to a function u in the RKHS is very friendly: it's clearly continuous and surjective. The problems arise in the inverse mapping: given a function in the manifold $\mathcal{M}^m$, trying to find coefficients and locations that produce that function is a multi-valued problem. This gets harder when u is viewed as an element of $\mathcal{M}^m$ but then also happens to belong to a smaller $\mathcal{M}^{m'}$ for $m' < m$, which is where the zero coefficients and coincident support locations arise, just as you mentioned.

However, for the sake of a continuous optimization, we need only the ability to differentiate through the forward mapping, which is never a problem, and need not concern ourselves with the troublesome inverse mapping. It's possible that, during early stages of an optimization, the method will 'jump' between different equivalent parts of the preimage $\delta^{-1}(\mathcal{M}^m)$, but this does not affect the convergence of the procedure, and we will add to the article a new figure illustrating this phenomenon. We will take advantage of the camera ready version to provide a new sub-section discussing these issues in further detail than can be accomplished here.

--

Thanks again for your effort and insights, and we look forward to a productive discussion period.

Response to Rebuttal: Revised Benchmarks and Remaining Concerns

Official Comment by Reviewer mDJr 28 Jul 2026, 21:41 (modified: 29 Jul 2026, 02:35)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer mDJr

Revisions (/revisions?id=yiZn9dZrO4)

Comment:

Thank you for the targeted non-sparse integral-functional ridge experiments. These experiments address my concern that the original ridge construction may be unusually aligned with L0MO's movable sparse-support parameterization, and they strengthen the empirical case for the method. Therefore, I remain weak acceptance but will raise confidence.

The reason I didn't raise the acceptance rating is that I do not think the manifold issue has been resolved. Continuity and surjectivity of the forward parameterization do not imply that $\mathcal{M}^m$ is globally a smooth $(d+1)m$-dimensional manifold. Zero coefficients, coincident supports, lower-sparsity representations, and boundary points form lower-dimensional or singular strata, and the pullback metric may become singular there. The statement that transitions between equivalent preimages "do not affect convergence" also requires justification. This isn't a fatal problem, but I think the paper's expression is mathematically inaccurate.

I also remain concerned about the hindsight hyperparameter protocol. The response asserts that rankings are mostly unaffected by the RKHS lengthscale but provides no quantitative evidence, and it does not appear to address the per-method/per-benchmark selection of the surrogate kernel profile. A ranking or sensitivity summary from the existing sweep, or an explicit narrowing of the claims to oracle-tuned upper-envelope performance, would be important.

Replying to Response to Rebuttal: Revised Benchmarks and Remaining Concerns

Official Comment by Authors

Official Comment

by Authors (Meghan E. Huber (/profile?id=~Meghan_E._Huber1), Donghyun Kim (/profile?id=~Donghyun_Kim11), Nathan Wycoff (/profile?id=~Nathan_Wycoff2), Davide Sartor (/profile?id=~Davide_Sartor1))

31 Jul 2026, 18:05

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer mDJr

Comment:

> Thank you for the targeted non-sparse integral-functional ridge experiments. These experiments address my concern that the original ridge construction may be unusually aligned with L0MO's movable sparse-support parameterization, and they strengthen the empirical case for the method. Therefore, I remain weak acceptance but will raise confidence.

We greatly appreciated your feedback on this point and we hope we can address your remaining concerns with this reply.

> The reason I didn't raise the acceptance rating is that I do not think the manifold issue has been resolved. Continuity and surjectivity of the forward parameterization do not imply that $\mathcal{M}^m$ is globally a smooth $(d+1)m$-dimensional manifold. Zero coefficients, coincident supports, lower-sparsity representations, and boundary points form lower-dimensional or singular strata, and the pullback metric may become singular there.

You are right, the statement in 172-178 is imprecise as presented. As you point out, the dimension of $\mathcal{M}^m$ is not locally constant. What we were after in that passage was a clear separation from prior work, which searches a fixed linear subspace.

Our search space is technically a stratified set, whose strata are the L^0 level sets

$\mathcal{S}^m = \{u \in \mathcal{H} : \|u\|_0 = m\}$

Each $\mathcal{S}^m$ is in fact a smooth manifold (for the rbf kernel we are considering): The canonical parameterization $\varphi(w) = \sum_{i=1}^m \alpha_i k(\cdot, x_i)$, $w = (\alpha, x)$, maps the open set $W^m = \{x_i \text{ pairwise distinct},\ \alpha_i \neq 0\} \subset \mathbb{R}^{(d+1)m}$ onto $\mathcal{S}^m$, $m!$-to-one. On W^m we have $G(w) \succ 0$, so $\mathcal{S}^m$ is an immersed smooth submanifold of $\mathcal{H}$ of dimension $(d+1)m$ and φ an $m!$-sheeted covering of it.

We would like to stress that the rest of the analysis remains valid, replacing "manifold" with "stratified set". You are also correct in pointing out that the metric G is degenerate outside of the top level strata. This is consistent with the metric of $\mathcal{H}$ (for instance, if $a_i = 0$ then any displacement in x_i has no effect on the norm). It does however change the correct preconditioning in Eq (8) from $G^{-1}\nabla f$ to $G^{+}\nabla f$, which is what we use in the code (see supplementary materials). We will make sure to rectify this.

This being said, the possibly singular nature of G is an aspect we did not explore with the required attention. To exclude the hypotesis that is hindering the optimization of the natural gradient variant, we have ran an additional ablation over the condition number of the pseudoinverse, discarding singular directions with $s_i < \texttt{rcond} \cdot s_1$. We find that truncation makes no practical difference.

rcond	None		1e-10		1e-08		1e-06	
	final	avg	final	avg	final	avg	final	avg
gramacylee [10^{-1}]	5.13	5.34	5.13	5.33	5.13	5.33	5.13	5.25
ackley [10^{1}]	0.68	1.05	0.57	1.02	0.66	1.01	0.65	1.09
hartmann [10^{0}]	1.77	2.26	1.75	2.31	1.87	2.33	1.69	2.32
rosenbrock [10^{4}]	4.16	5.29	4.43	5.35	4.17	5.31	4.19	5.27
pendulum [10^{2}]	3.98	4.63	3.96	4.62	3.95	4.58	3.90	4.54
brachistochrone [10^{-2}]	2.11	5.34	2.09	5.51	1.87	5.39	1.78	5.21

> The statement that transitions between equivalent preimages “do not affect convergence” also requires justification.

We tested the impact of sheet boundary transitions directly, comparing the naive $m!$-to-one parameterization against a bijective one which searches a single sheet (support points sorted, with relative coordinates). There is no statistically significant difference between the two on any configurations. Theoretically this is what one expects, since the acquisition is permutation invariant and the $m!$ sheets carry identical landscapes. We report a summary of the results below and will include this study in the appendix.

	Multi Sheet		Single Sheet	
	final	avg	final	avg
gramacylee [10^{-1}]	5.13	5.34	5.13	5.33
ackley [10^{1}]	0.68	1.05	0.61	1.03
hartmann [10^{0}]	1.77	2.26	1.70	2.28
rosenbrock [10^{4}]	4.16	5.29	4.33	5.33
pendulum [10^{2}]	3.98	4.63	4.03	4.66
brachistochrone [10^{-2}]	2.11	5.34	2.17	5.49

Official Comment by Authors

Official Comment

by Authors (Meghan E. Huber (/profile?id=~Meghan_E._Huber1), Donghyun Kim (/profile?id=~Donghyun_Kim11), Nathan Wycoff (/profile?id=~Nathan_Wycoff2), Davide Sartor (/profile?id=~Davide_Sartor1))

31 Jul 2026, 18:10 (modified: 31 Jul 2026, 18:11)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer mDJr

Revisions (/revisions?id=yQH7loXHPG)

Comment:

> I also remain concerned about the hindsight hyperparameter protocol. The response asserts that rankings are mostly unaffected by the RKHS lengthscale but provides no quantitative evidence, and it does not appear to address the per-method/per-benchmark selection of the surrogate kernel profile. A ranking or sensitivity summary from the existing sweep, or an explicit narrowing of the claims to oracle-tuned upper-envelope performance, would be important.

We would like to specify while we select a shared lenghtscale for the comparison, the profile is method dependent. In Appendix A we already show the sensitifity of the results on the choice of kernel profile. As we point out, the results are not particularly sensitive to this parameter, with Matérn 3/2 or Matérn 5/2 both being solid defaults, similarly to tranditional BO.

The following are the results fixing the Matérn 5/2 profile for all methods. Ours is 1st (or tied for 1st) on all benchmarks.

Problem	**Random**		**Ours**		**Kundu**		**Vien**		**Shilton**		**Vellanki**	
	final	avg	final	avg	final	avg	final	avg	final	avg	final	avg
gramacylee [10^{-1}]	5.19	**5.53**	**5.13**	**5.39**	6.37	6.58	5.13	**5.45**	**5.13**	**5.50**	**5.13**	5.46
ackley [10^{1}]	1.46	1.58	**0.09**	**0.66**	1.51	1.60	1.58	1.61	1.16	1.39	N/A	N/A
hartmann [10^{0}]	3.04	3.14	**0.78**	**1.36**	2.65	2.94	2.38	2.55	2.43	2.58	N/A	N/A
rosenbrock [10^{4}]	6.10	6.57	**0.66**	**3.17**	5.33	5.91	5.10	5.51	4.43	5.07	N/A	N/A
pinwheel [10^{0}]	1.26	1.95	**0.12**	**1.07**	1.55	1.79	2.78	3.17	1.54	1.72	2.54	2.82
pendulum [10^{2}]	4.99	5.29	**3.43**	**4.31**	4.95	5.09	4.75	5.15	4.34	4.74	N/A	N/A
hopper [10^{0}]	-2.36	-1.84	**-2.53**	**-2.16**	-1.89	-1.86	-1.54	-1.50	**-2.51**	**-2.23**	-1.47	-1.46
brachistochrone [10^{-1}]	0.98	1.24	**0.10**	**0.47**	0.74	1.21	0.38	0.74	**0.10**	**0.52**	0.23	0.91
mnist [10^{-2}]	2.16	2.18	**2.13**	**2.15**	2.19	2.28	2.15	2.18	**2.14**	2.18	2.18	2.34
avg. rank	*3.89*	*3.67*	*1.00*	*1.00*	*4.33*	*4.00*	*3.67*	*3.22*	*1.67*	*1.67*	*3.80*	*4.40*

We did not report a full sensitivity analisys for each competitor as we are comparing to the *best profile*. Any sensitivity of competitors to this hyperparameter only plays in our favor, making our results even more significant.

We would also like to substantiate our claim that "rankings are mostly unaffected by the RKHS lengthscale". The following table reports the mean rank over problems at each lengthscale. Our method is the clear favourite regardless of lenghtscale.

rho	**Random**		**Ours**		**Kundu**		**Vien**		**Shilton**		**Vellanki**	
	final	avg	final	avg	final	avg	final	avg	final	avg	final	avg
0.05	4.67	4.33	**1.78**	**2.00**	3.11	3.67	2.44	2.89	2.89	2.33	3.20	3.20
0.1	4.33	4.11	**1.67**	**1.78**	3.22	3.33	3.00	3.22	2.56	2.67	3.20	3.20
0.2	4.33	4.22	**1.11**	**1.11**	3.67	3.89	3.56	3.22	2.33	2.22	3.40	3.40

rho	Random		Ours		Kundu		Vien		Shilton		Vellanki	
0.4	3.67	3.33	**1.11**	**1.11**	4.22	4.11	3.67	3.22	2.33	1.78	4.00	4.60

We believe that hindsight selection is a more interesting comparison, since if the leghtscale is choosen poorly, no good solution exists, so all methods perform poorly in the absolute sense and being "the best" relatively is not particularly meaningful.

We will make sure to include this comparison in the camera ready version, as well as the results for each individual lenghtscale and profile combination that we omit here due to space constraints.

Replying to Official Comment by Authors

The manifold expression has been clarified.

Official Comment by Reviewer mDJr 02 Aug 2026, 00:37

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer mDJr

Comment:

Thank you for the follow-up response. I believe the paper now corrects the central geometric issue in the appropriate direction. In particular, explicitly acknowledge that the at-most-m sparse set is (typically) a single globally smooth manifold, but rather a stratified union of $\binom{n}{m}$ sparsity components. The new pseudoinverse and parameterization ablations also suggest that the singular and permutation-redundant regions do not materially affect the numerical performance of the optimizer. I believe the paper is now sound in its expression, therefore I continue to recommend acceptance.

About OpenReview (/about) Contact (/contact)
FAQ (https://docs.openreview.net/getting-started/frequently-asked-questions) Hosting a Venue (/group?id=OpenReview.net/Support)
Sponsors (/sponsors) Terms of Use (/legal/terms) Privacy Policy (/legal/privacy)
All Venues (/venues) **Donate** (/donate)
OpenReview (/about) is a long-term project to advance science through improved peer review with legal nonprofit status. We gratefully acknowledge the support of the OpenReview Sponsors (/sponsors). © 2026 OpenReview

Replying to The manifold expression has been clarified.

Official Comment by Authors

Official Comment

by Authors (Meghan E. Huber (/profile?id=~Meghan_E._Huber1), Donghyun Kim (/profile?id=~Donghyun_Kim11), Nathan Wycoff (/profile?id=~Nathan_Wycoff2), Davide Sartor (/profile?id=~Davide_Sartor1))

03 Aug 2026, 22:32

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer mDJr

Comment:

Thanks for confirming that the geometric statement is now sound, and for the recommendation.

Your review helped us materially improve the solidity of the results. Thank you.